



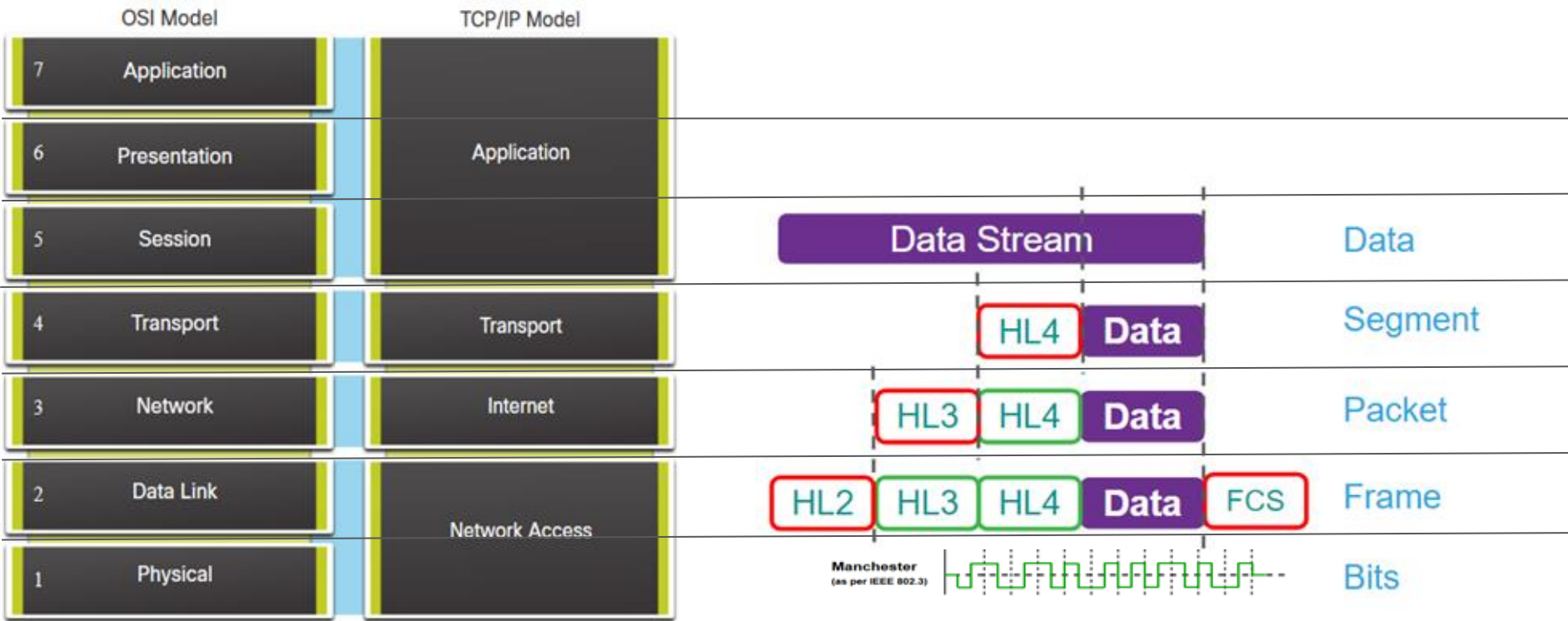
Module 4: Physical Layer

Telecom Academy
Cristian Deacu

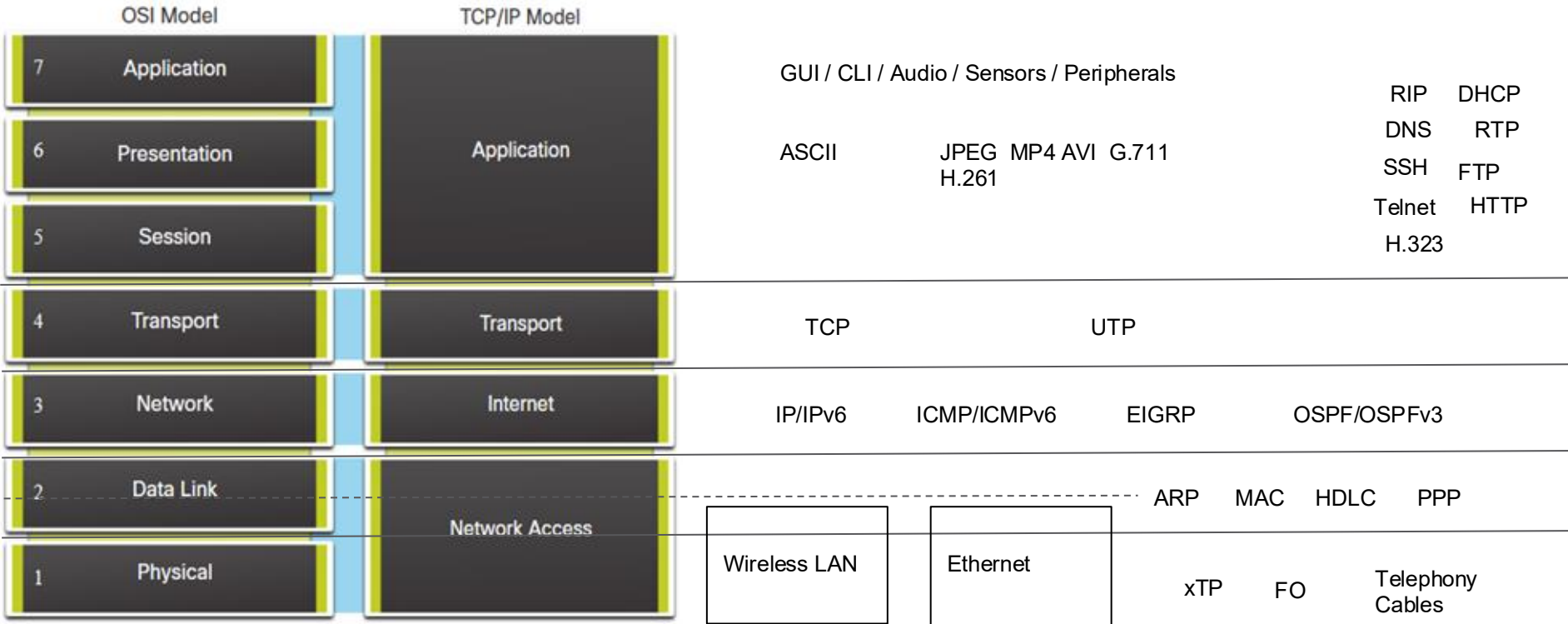
Introduction to Networks v7.0
(ITN)



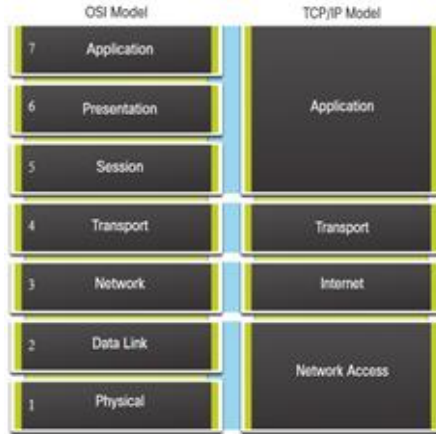
Cheat Sheet



Cheat Sheet



Specialized Jobs



4.1 Purpose of the Physical Layer



Purpose of the Physical Layer

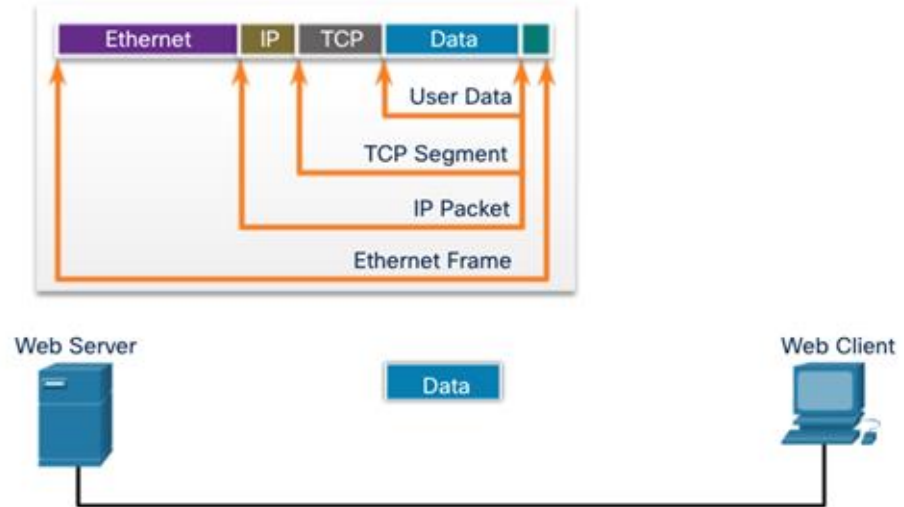
The Physical Layer

Transports bits across the network media

Accepts a complete frame from the Data Link Layer and encodes it as a series of signals that are transmitted to the local media

This is the last step in the encapsulation process.

The next device in the path to the destination receives the bits and re-encapsulates the frame, then decides what to do with it.

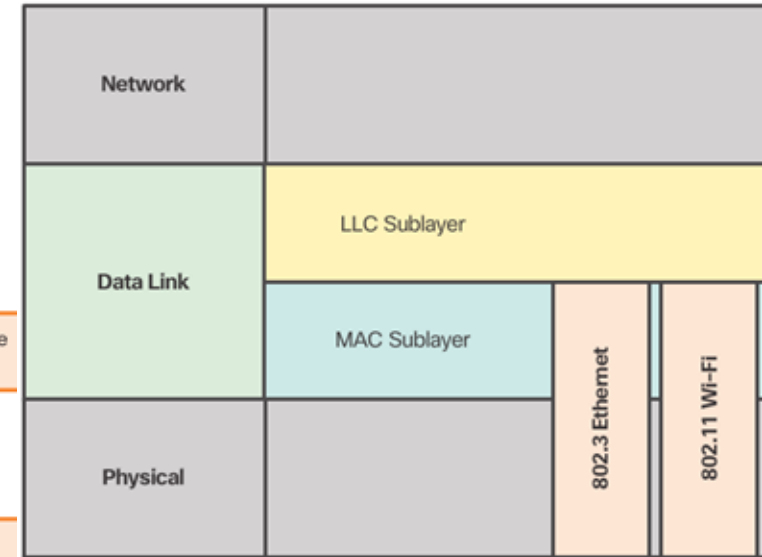
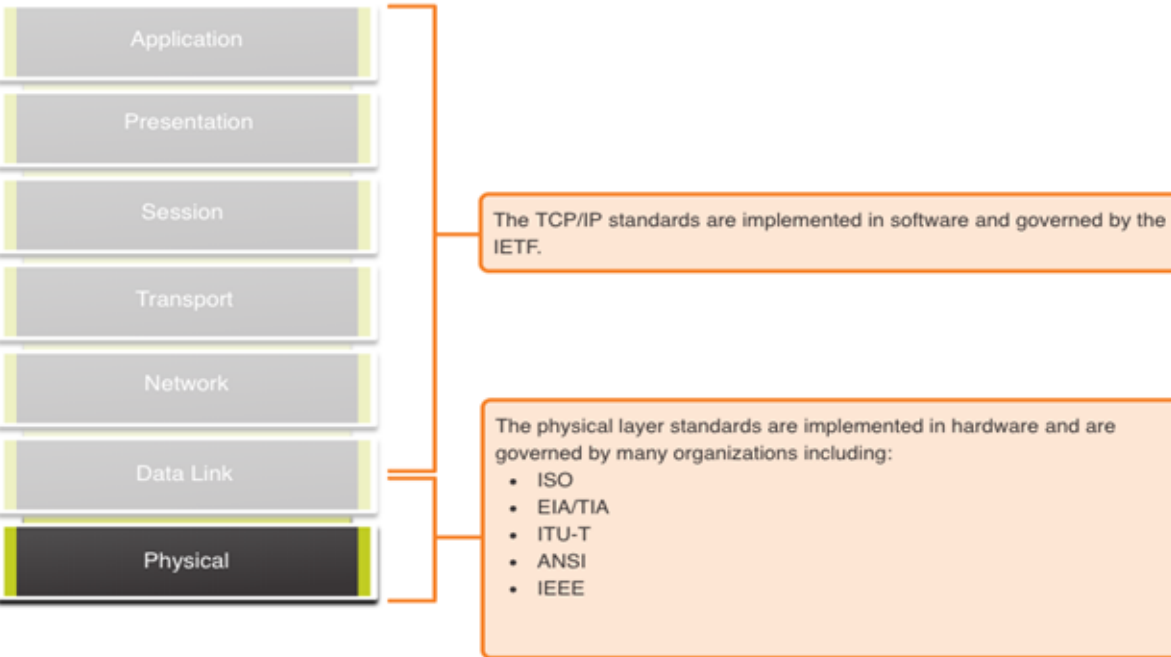


4.2 Physical Layer Characteristics



Physical Layer Characteristics

Physical Layer Standards



Physical Layer Characteristics

Physical Components

Physical Layer Standards address three functional areas:

- Physical Components
- Encoding
- Signaling

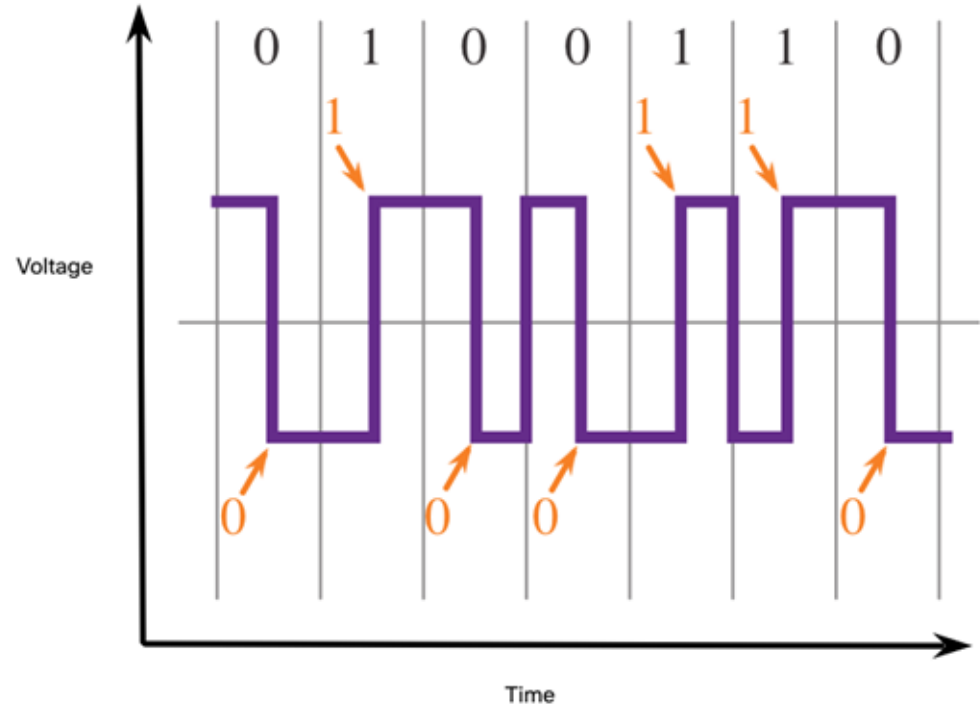
The Physical Components are the hardware devices, media, and other connectors that transmit the signals that represent the bits.

- Hardware components like NICs, interfaces and connectors, cable materials, and cable designs are all specified in standards associated with the physical layer.

Physical Layer Characteristics

Encoding

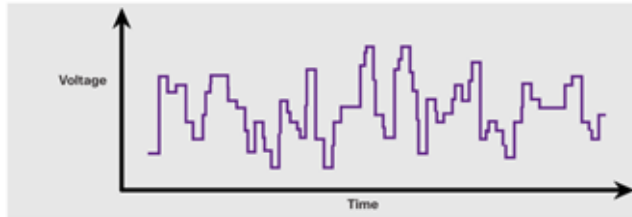
- Encoding converts the stream of bits into a format recognizable by the next device in the network path.
- This 'coding' provides predictable patterns that can be recognized by the next device.
- Examples of encoding methods include Manchester (shown in the figure), 4B/5B, and 8B/10B.



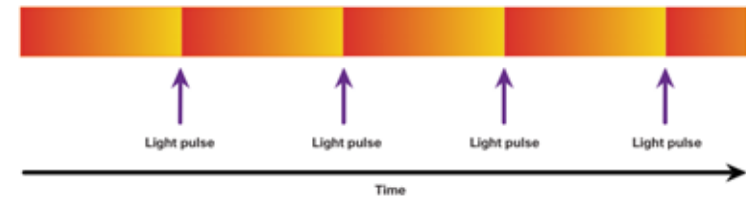
Physical Layer Characteristics

Signaling

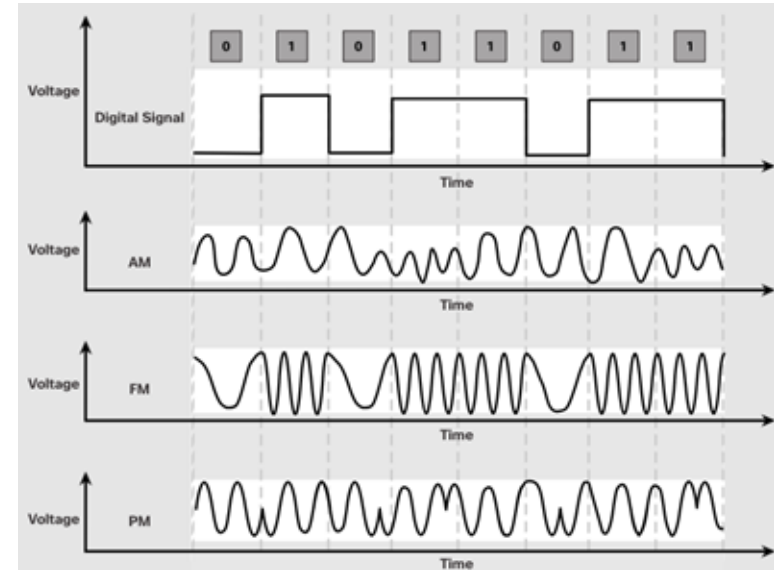
- The signaling method is how the bit values, “1” and “0” are represented on the physical medium.
- The method of signaling will vary based on the type of medium being used.



Electrical Signals Over Copper Cable



Light Pulses Over Fiber-Optic Cable



Microwave Signals Over Wireless

Physical Layer Characteristics

Bandwidth

- Bandwidth is the capacity at which a medium can carry data.
- Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time; how many bits can be transmitted in a second.
- Physical media properties, current technologies, and the laws of physics play a role in determining available bandwidth.

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

Physical Layer Characteristics

Bandwidth Terminology

Latency

- Amount of time, including delays, for data to travel from one given point to another

Throughput

- The measure of the transfer of bits across the media over a given period of time

Goodput

- The measure of usable data transferred over a given period of time
- $\text{Goodput} = \text{Throughput} - \text{traffic overhead}$

4.3 Copper Cabling



Copper Cabling

Types of Copper Cabling



Unshielded Twisted-Pair (UTP) Cable



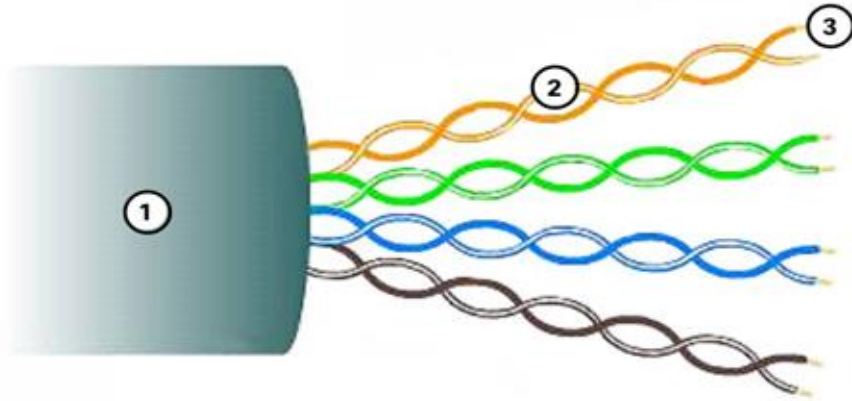
Shielded Twisted-Pair (STP) Cable



Coaxial Cable

Copper Cabling

Unshielded Twisted Pair (UTP)



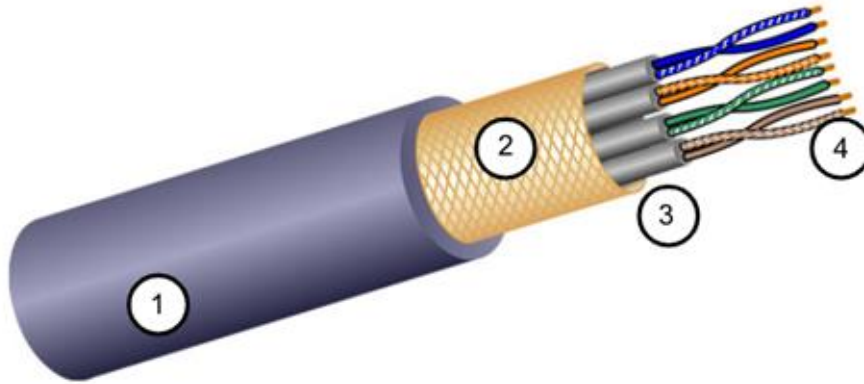
- UTP is the most common networking media.
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices.

Key Characteristics of UTP

1. The outer jacket protects the copper wires from physical damage.
2. Twisted pairs protect the signal from interference.
3. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair.

Shielded Twisted Pair (STP) Shielded Foiled Twisted Pair (SFTP)

- Better noise protection than UTP
- More expensive than UTP
- Harder to install than UTP
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices



Key Characteristics of STP

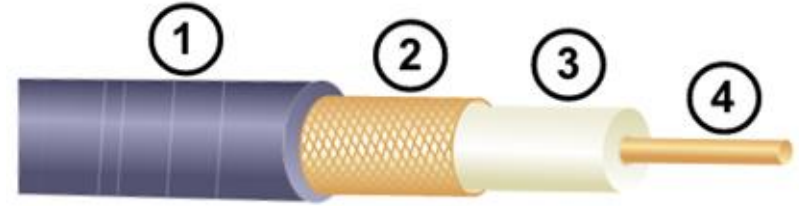
1. The outer jacket protects the copper wires from physical damage
2. Braided or foil shield provides EMI/RFI protection
3. Foil shield for each pair of wires provides EMI/RFI protection
4. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair

Copper Cabling

Coaxial Cable

Consists of the following:

1. Outer cable jacket to prevent minor physical damage
2. A woven copper braid, or metallic foil, acts as the second wire in the circuit and as a shield for the inner conductor.
3. A layer of flexible plastic insulation
4. A copper conductor is used to transmit the electronic signals.



There are different types of connectors used with coax cable.

Commonly used in the following situations:

- Wireless installations - attach antennas to wireless devices
- Cable internet installations - customer premises wiring



4.4 UTP Cabling



Properties of UTP Cabling

UTP has four pairs of color-coded copper wires twisted together and encased in a flexible plastic sheath. No shielding is used. UTP relies on the following properties to limit crosstalk:

- Cancellation - Each wire in a pair of wires uses opposite polarity. One wire is negative, the other wire is positive. They are twisted together and the magnetic fields effectively cancel each other and outside EMI/RFI.
- Variation in twists per foot in each wire - Each wire is twisted a different amount, which helps prevent crosstalk amongst the wires in the cable.



UTP Cabling Standards and Connectors

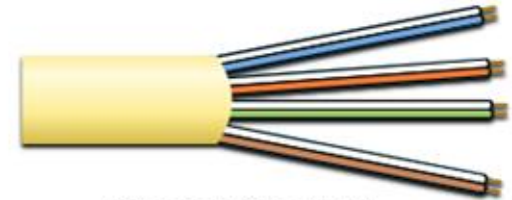
Standards for UTP are established by the TIA/EIA. TIA/EIA-568 standardizes elements like:

- Cable Types
- Cable Lengths
- Connectors
- Cable Termination
- Testing Methods

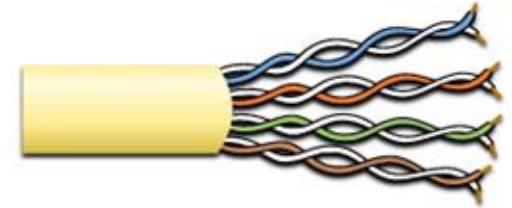
Electrical standards for copper cabling are established by the IEEE, which rates cable according to its performance.

Examples include:

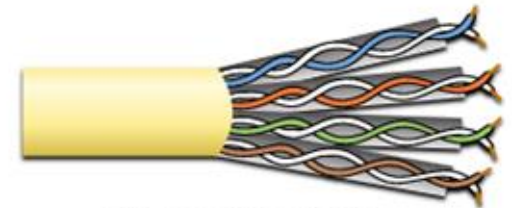
- Category 3
- Category 5 and 5e
- Category 6/6a - 7



Category 3 Cable (UTP)



Category 5 and 5e Cable (UTP)



Category 6 Cable (UTP)

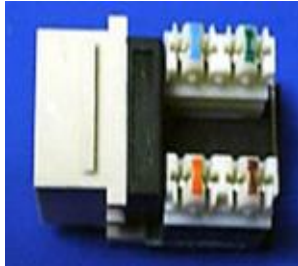
UTP Cabling Standards and Connectors (Cont.)



RJ-45 Connector



Poorly terminated UTP cable



RJ-45 Socket

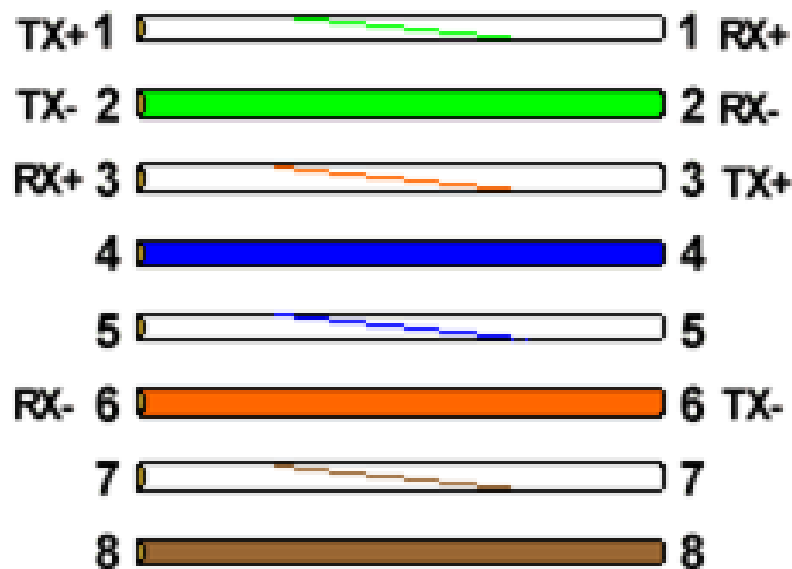


Properly terminated UTP cable

Tipuri de cabluri

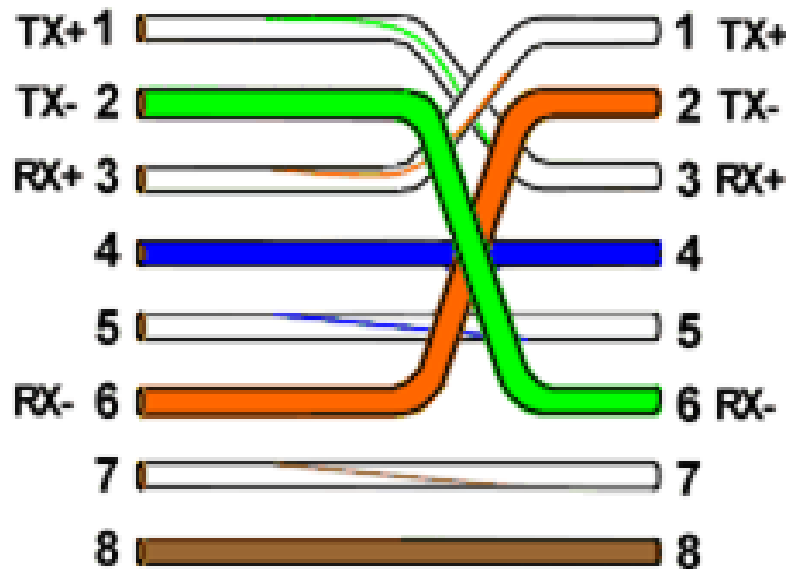
Router
PC

Switch
AP



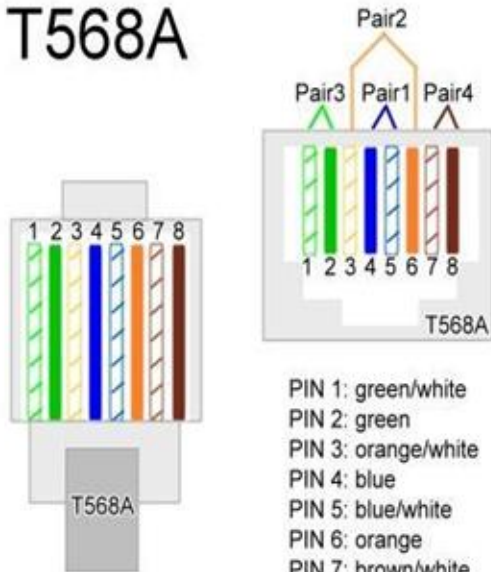
Router
PC

Router
PC

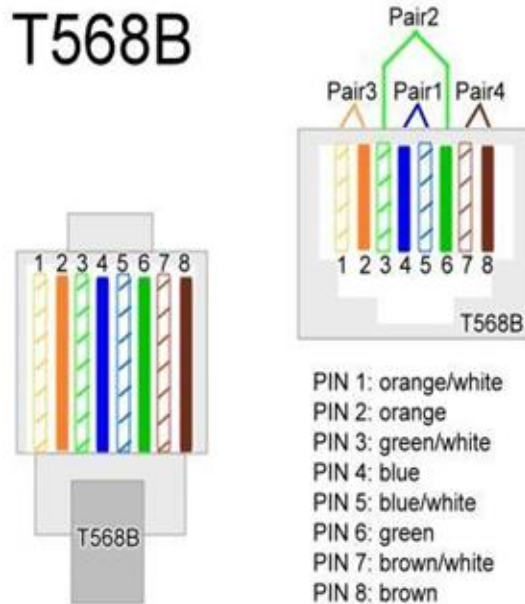


LAB Sertizare

T568A



T568B



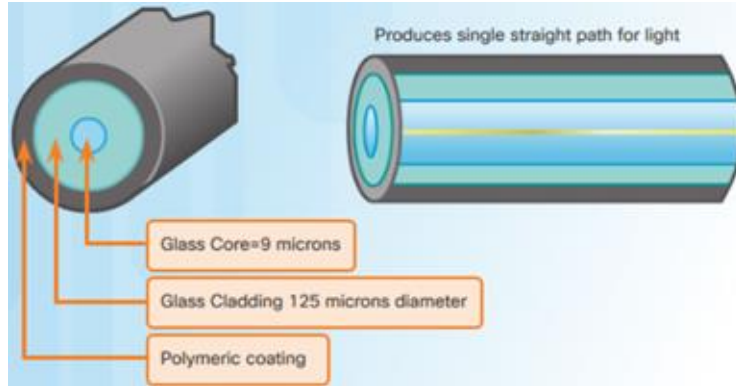
4.5 Fiber-Optic Cabling



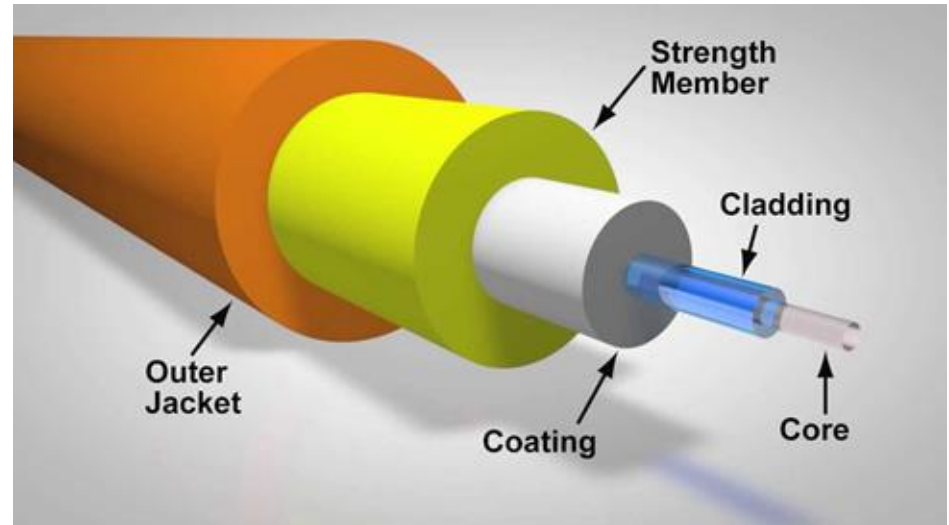
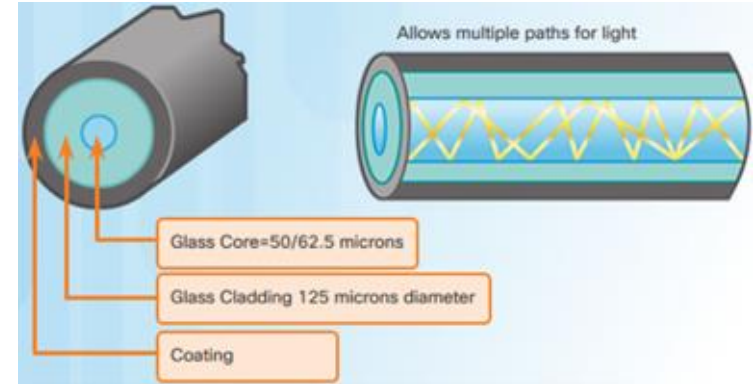
Fiber-Optic Cabling

Types of Fiber Media

Single-Mode Fiber



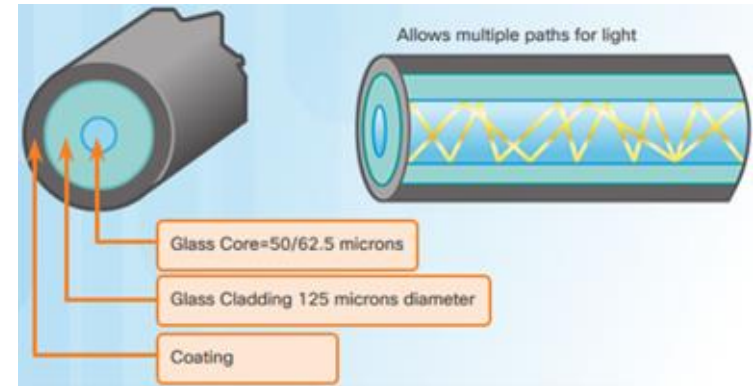
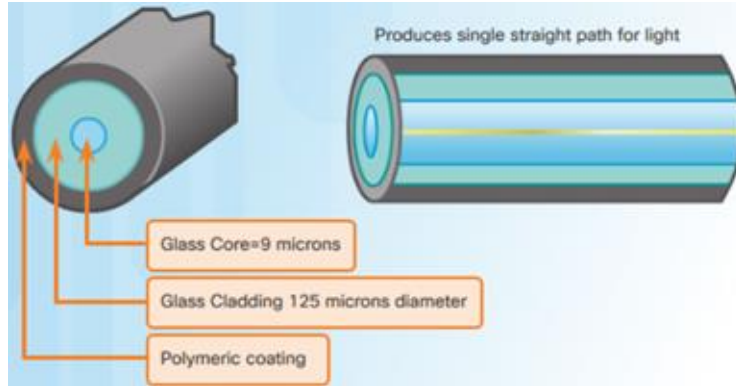
- Very small core
- Uses expensive lasers
- Long-distance applications



Fiber-Optic Cabling

Types of Fiber Media

Single-Mode Fiber



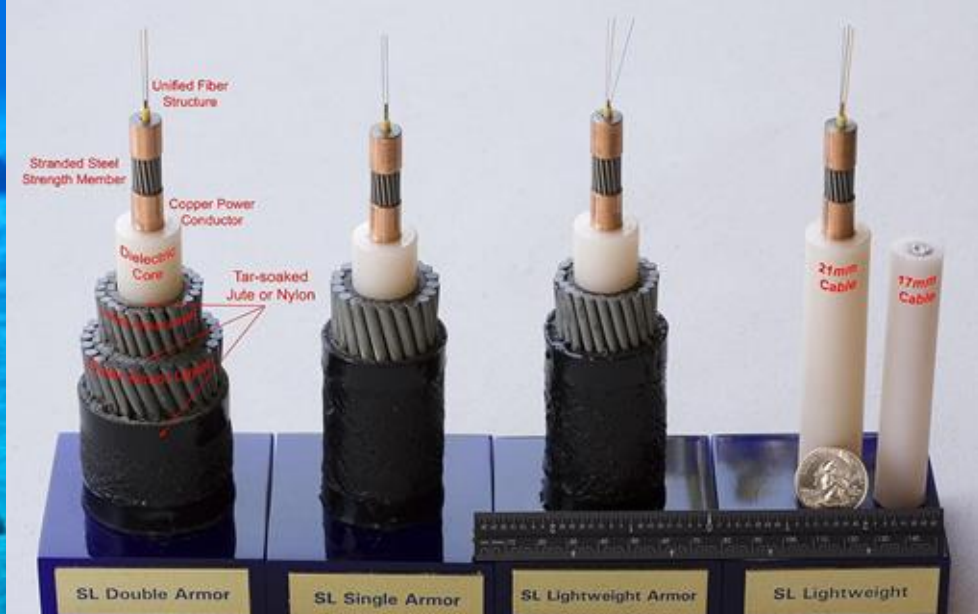
Dacă $V < 2.405$, fibra este **monomodă**, adică poate transmite doar un singur mod de lumină.

Dacă $V > 2.405$, fibra va fi **multimodă**, adică poate transporta mai multe moduri de lumină simultan.

$$V = \frac{2\pi a}{\lambda} \cdot NA$$







Fiber-Optic Cabling

Fiber-Optic Connectors



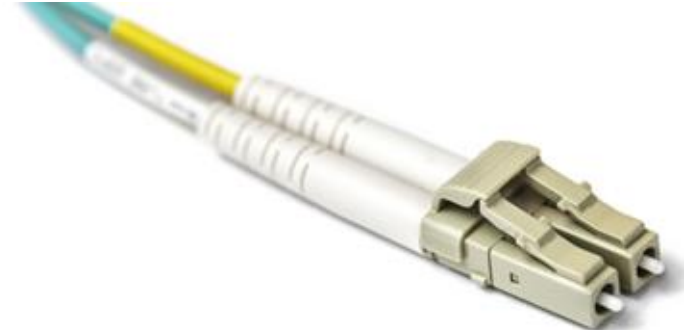
Straight-Tip (ST) Connectors



Lucent Connector (LC) Simplex Connectors



Subscriber Connector (SC) Connectors



Duplex Multimode LC Connectors

Fiber-Optic Cabling

Fiber Patch Cords



SC-SC MM Patch Cord



LC-LC SM Patch Cord



ST-LC MM Patch Cord



ST-SC SM Patch Cord

A yellow jacket is for single-mode fiber cables and orange (or aqua) for multimode fiber cables.

Fiber-Optic Cabling

Fiber versus Copper

Optical fiber is primarily used as backbone cabling for high-traffic, point-to-point connections between data distribution facilities and for the interconnection of buildings in multi-building campuses.

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 10 Gb/s	10 Mb/s - 100 Gb/s
Distance	Relatively short (1 - 100 meters)	Relatively long (1 - 100,000 meters)
Immunity to EMI and RFI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest

4.6 Wireless Media



Wireless Media

Properties of Wireless Media

Coverage area - Effective coverage can be significantly impacted by the physical characteristics of the deployment location.

- **Interference**
- **Security**
- **Shared medium** - in half-duplex

Wireless Standards:

Wi-Fi (IEEE 802.11)

Bluetooth (IEEE 802.15)

WiMAX (IEEE 802.16)

Zigbee (IEEE 802.15.4)



Version	WiFi 4	WiFi 5	WiFi 6	WiFi 6E	WiFi 7
Year	2007	2013	2019	2020	2024
Protocol	802.11n	802.11ac	802.11ax	802.11ax	802.11be
Bands	2.4 / 5 GHz	5 GHz only	2.4 / 5 GHz	2.4 / 5 / 6* GHz	2.4 / 5 / 6 GHz
Peak Speed	600 Mb/s	6.9 Gb/s	9.6 Gb/s	9.6 Gb/s	46 Gb/s
Channel Widths	20 / 40 MHz	20 / 40 / 80 / 160 MHz	20 / 40 / 80 / 160 MHz	20 / 40 / 80 / 160 MHz	20 / 40 / 80 / 160 / 320 MHz
Security	WPA2	WPA2	WPA3	WPA3	WPA3
Key Features	4 x 4 MIMO	8 x 8 MIMO	8 x 8 MIMO	8 x 8 MIMO	16 x 16 MIMO
	LDPC Error Correction	4 x DL MU-MIMO	8 x DL MU-MIMO	8 x DL MU-MIMO	16 x DL & UL MU-MIMO
	64-QAM	Beam Forming*	Beam Forming	Beam Forming	Multi-AP
		256-QAM	OFDMA	OFDMA	Multi-RU Puncturing
			TWT	TWT	Multi-Link
			1024-QAM	1024-QAM	Beam Forming
				*6 GHz added, USA Only	OFDMA
					TWT
					4096-QAM

16x DL & UL MU-MIMO - Multi-User MIMO cu 16 fluxuri de date pentru **Downlink (DL)** și **Uplink (UL)**

Multi-AP permite **colaborarea între mai multe puncte de AP-uri**: acoperire, frecvențe, roaming, traffic management, scalabilitate.

TWT (Target Wake Time) caracteristică a Wi-Fi 6 și Wi-Fi 7 care permite dispozitivelor să economisească energie prin sincronizarea orelor în care trebuie să transmită sau să primească date. (aplicație în IoT)

OFDMA (Orthogonal Frequency Division Multiple Access) - transmisiuni simultane pe același canal. +

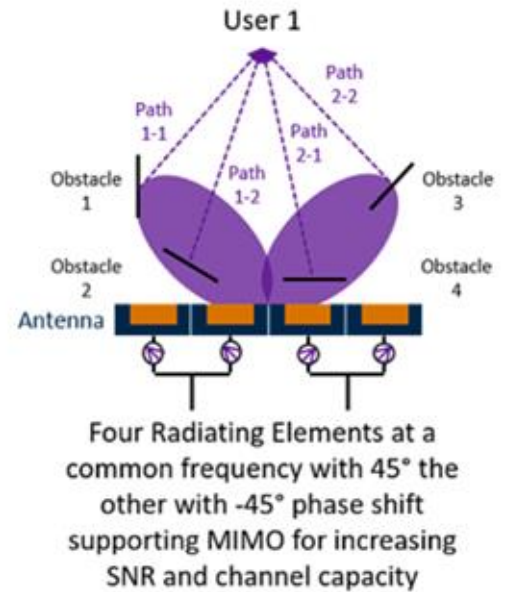
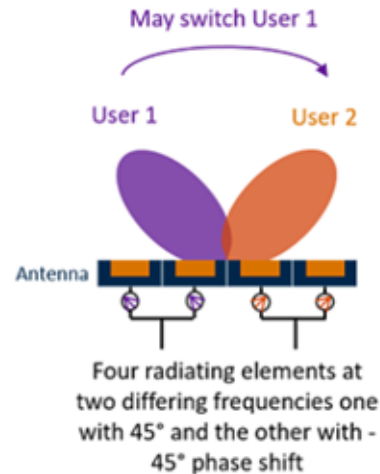
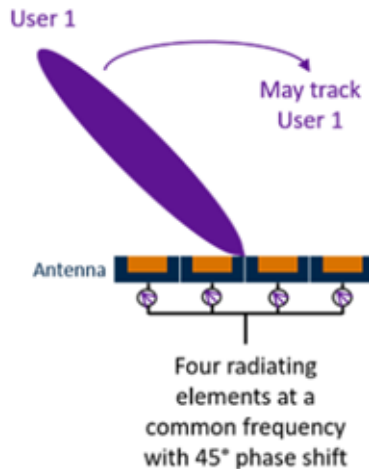
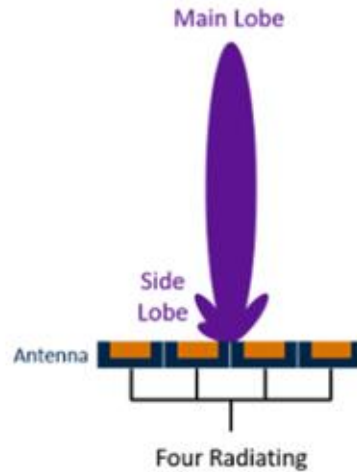
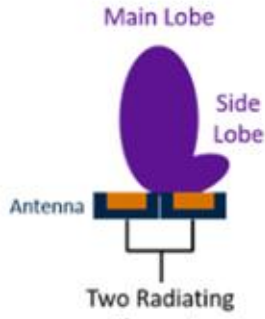
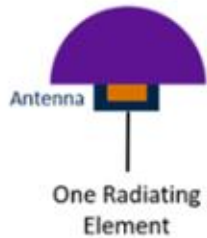
Multi-Link Operation (MLO) este o caracteristică a **Wi-Fi 7 (802.11be)** care permite unui dispozitiv să utilizeze simultan **mai multe benzi de frecvență** (de exemplu, 2.4 GHz, 5 GHz și 6 GHz)

1024-QAM poate transmite 10 biți pe simbol
4096-QAM poate transmite 12 biți pe simbol
 (combinație unică de amplitudine și fază)

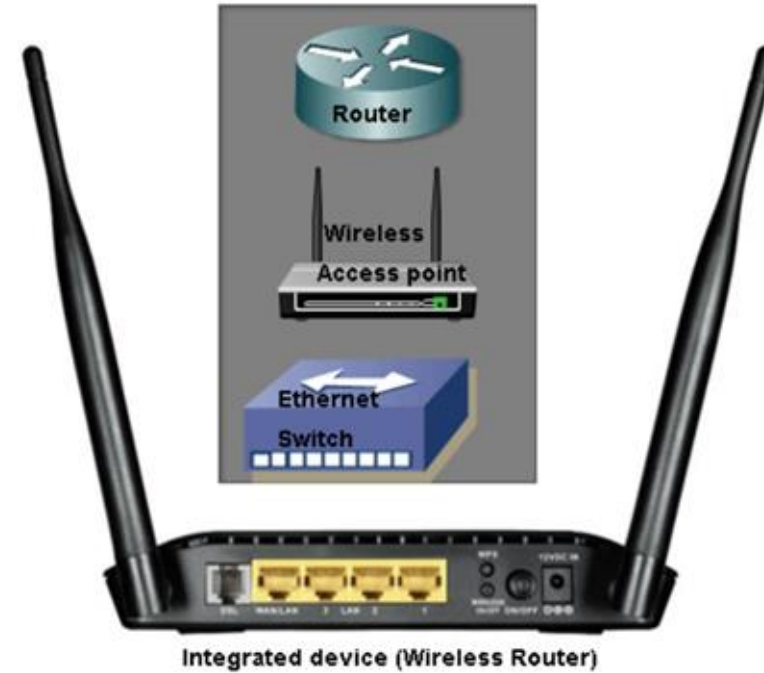
Multi-RU puncturing În **Wi-Fi 6** și **Wi-Fi 7**, canalele de frecvență mari (de exemplu, 80 MHz sau 160 MHz) sunt împărțite în **unități de resurse (RU)**. Fiecare unitate de resurse (RU) reprezintă un bloc de frecvență în care sunt transmise date.

WPA2 și **WPA3**

- criptare: AES vs GHASH
- Autentificare: PSK vs SAE
- criptare după chei individuale



LAB Router Wireless



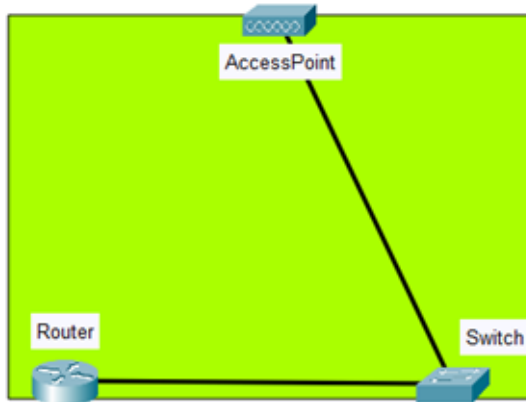
SOHO Router AP

Router+Switch+Access Point

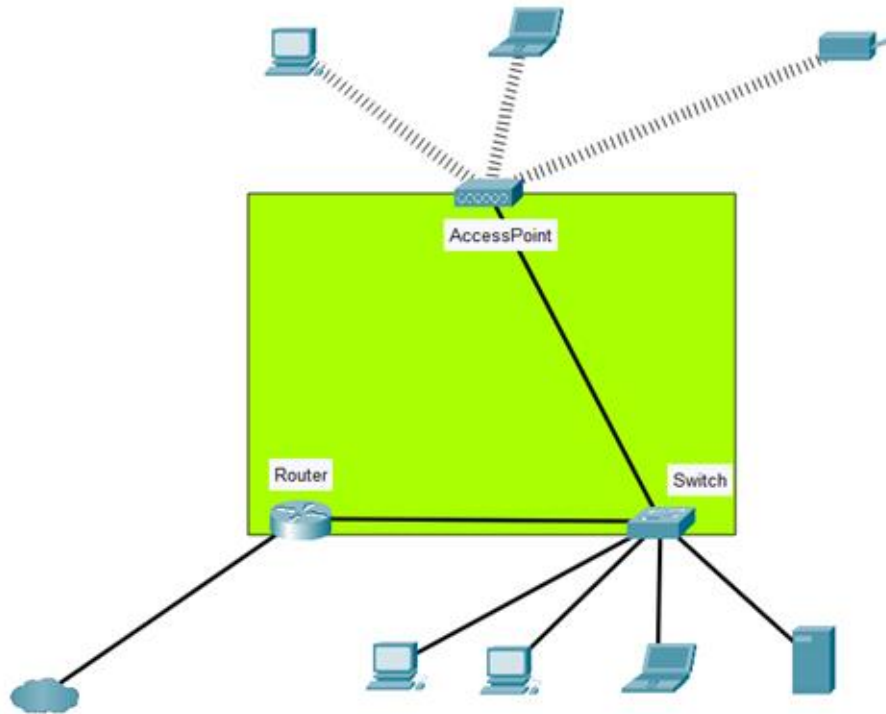


SOHO Router AP

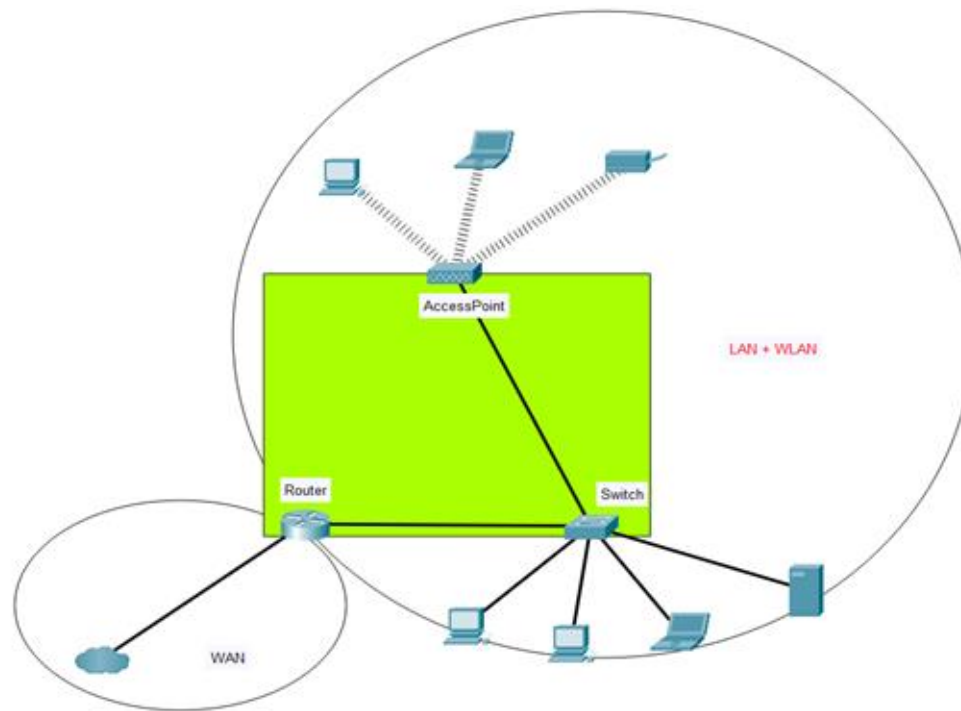
Router+Switch+Access Point



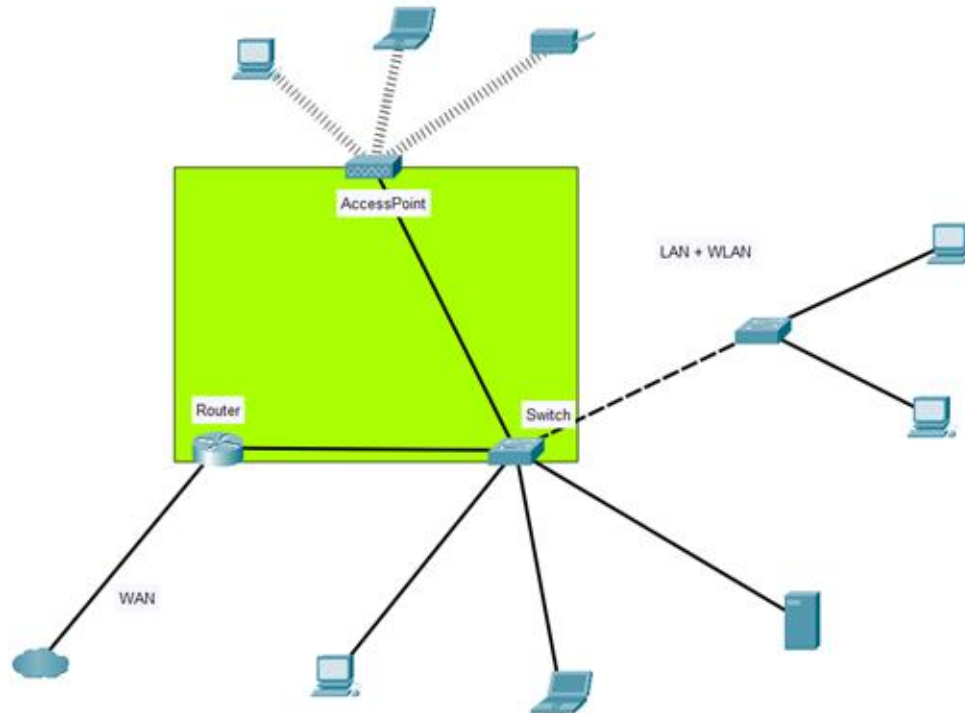
Routerul Separă Rețelele și Oferă Servicii



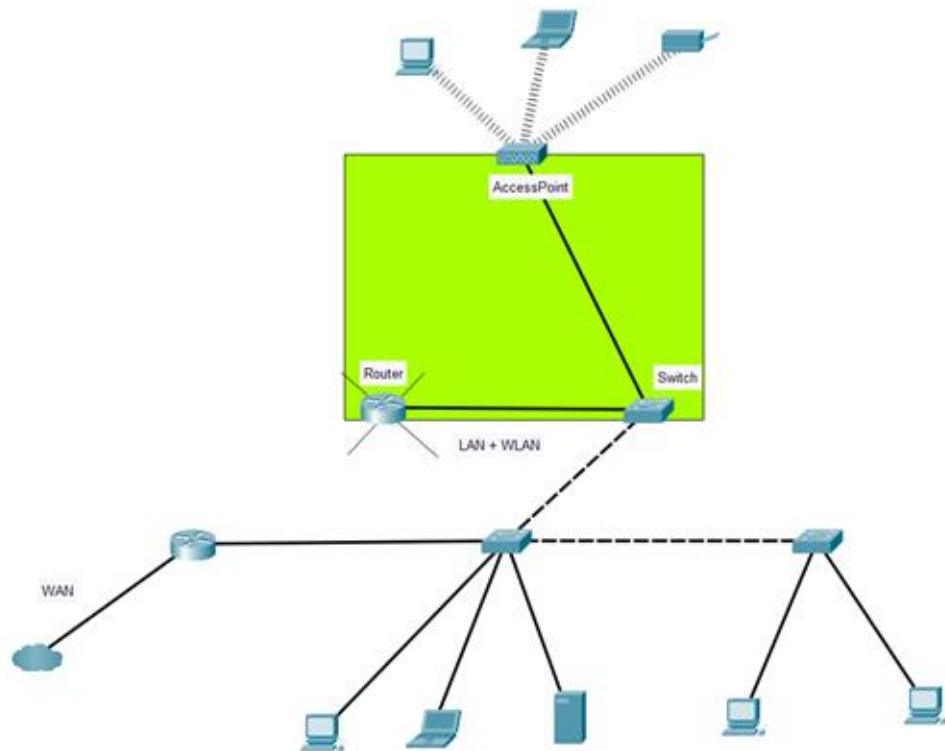
Routerul Separă Rețelele și Oferă Servicii



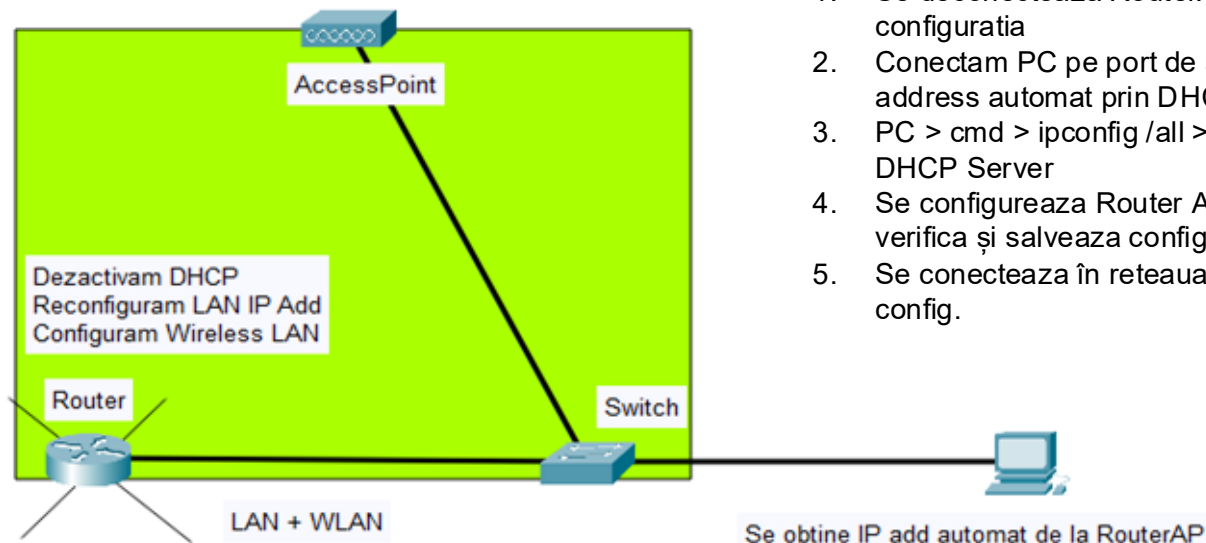
Scenariul 1. Home Office



Scenariul 2. Small Office



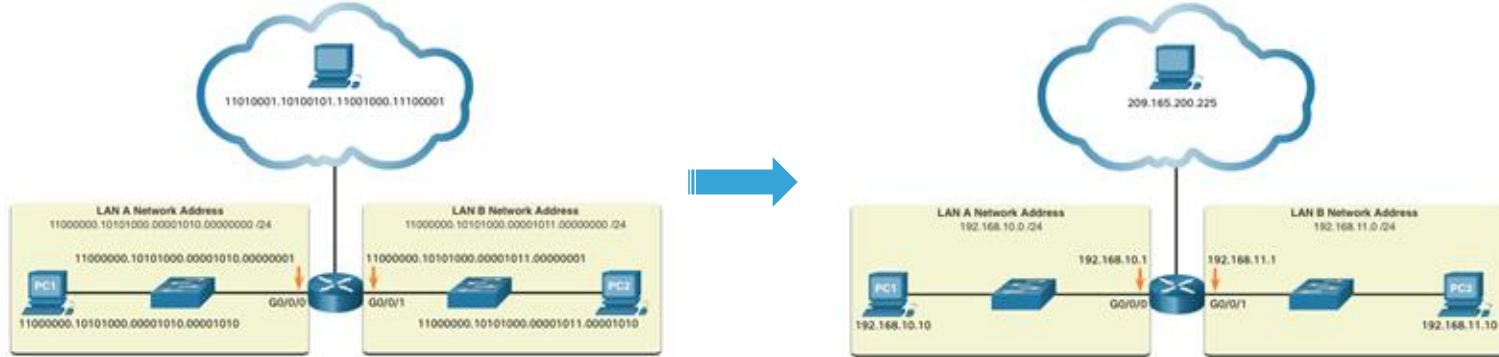
Configurare Scenariu AP



1. Se deconecteaza RouterAP și se reseteaza configuratia
2. Conectam PC pe port de switch și setam obtinerea IP address automat prin DHCP
3. PC > cmd > ipconfig /all > se copiaza în browser IP add DHCP Server
4. Se configureaza Router AP în functie de scenariu. Se verifica și salveaza config.
5. Se conecteaza în rețeaua din "productie". Se verifica config.

Binary Number System

Binary and IPv4 Addresses



Binary Number System

Binary Positional Notation

Positional notation means that a digit represents different values depending on the “position” the digit occupies in the sequence of numbers.

The decimal positional notation system operates as shown in the tables below.

Radix	10	10	10	10
Position in Number	3	2	1	0
Calculate	(10^3)	(10^2)	(10^1)	(10^0)
Position Value	1000	100	10	1



	Thousands	Hundreds	Tens	Ones
Positional Value	1000	100	10	1
Decimal Number (1234)	1	2	3	4
Calculate	1×1000	2×100	3×10	4×1
Add them up...	1000	+ 200	+ 30	+ 4
Result	1,234			



Binary Number System

Binary Positional Notation (Cont.)

The binary positional notation system operates as shown in the tables below.

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position Value	128	64	32	16	8	4	2	1



Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							



Binary Number System

Convert Binary to Decimal

Convert 11000000.10101000.00001011.00001010 to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	1x128	0x64	1x32	0x16	1x8	0x4	0x2	0x1
Add Them Up...	128	+ 0	+ 32	+ 0	+ 8	+ 0	+ 0	+ 0
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	1x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 1
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	0x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 0

➡ 192

➡ 168

➡ 1

1

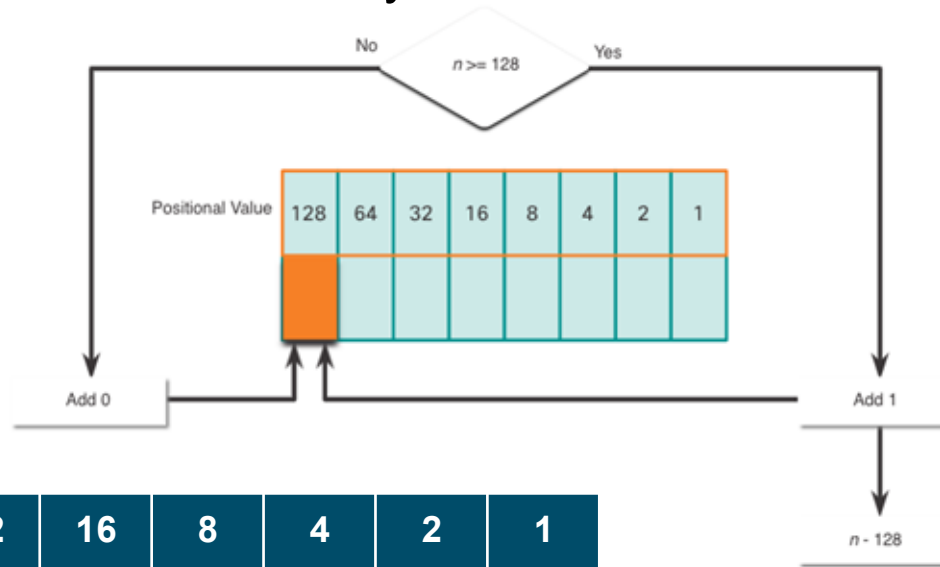
➡ 10

192.168.11.10

Binary Number System

Decimal to Binary Conversion

Convert decimal 168 to binary



128	64	32	16	8	4	2	1

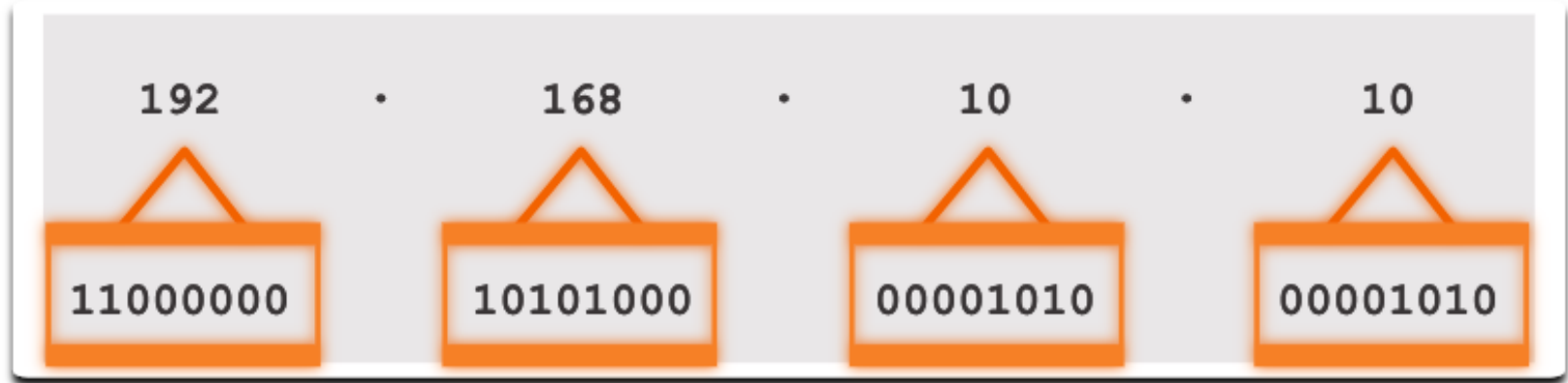
Binary Number System

IPv4 Addresses



Binary Number System

IPv4 Addresses



5.2 Hexadecimal Number System



Hexadecimal Number System

Hexadecimal and IPv6 Addresses

Decimal
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Binary
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111

Hexadecimal
0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F

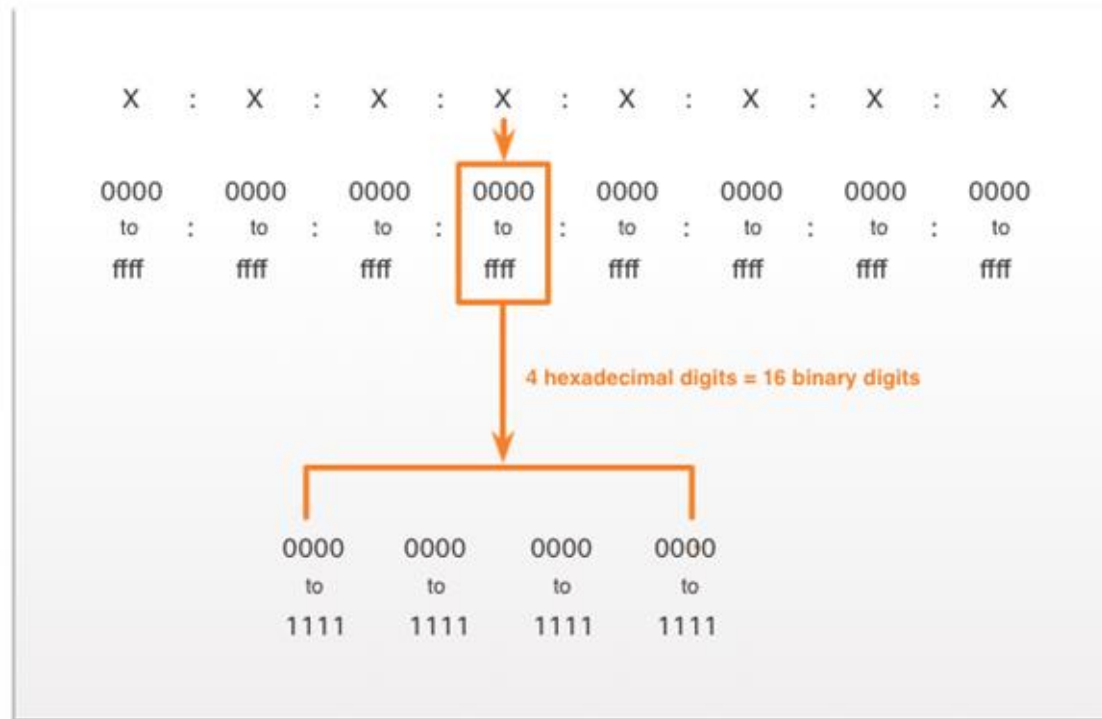


Seminar Transformări

$2 \leftrightarrow 10 \leftrightarrow 16$



Hexadecimal and IPv6 Addresses (Cont.)



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

_____	_____	_____	_____	_____	_____	_____	_____
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal --- Metode?!



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$201 - 128 =$$

—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

3. Binar > Hexazecimal

0101.1011₂

Deci	Heza	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal-Hexazecimal

3. Binar > Hexazecimal

0101.1011₂
5 B₁₆

4. Binar < Hexazecimal

A 6₁₆

Deci	Heza	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal-Hexazecimal

5. Zecimal > Binar > Hexazecimal

100

Deci	Heza	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal-Hexazecimal

5. Zecimal > Hexazecimal --- Metode?!



Transformări Binar-Zecimal-Hexazecimal

6. Zecimal < Binar < Hexazecimal

C

4

Deci	Heza	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



